

GOVT 6019: Estimation and Large Samples

Fall 2026

Melissa Sands (Instructor)

Izza Malik (Teaching Assistant)

1. Tools

1.1 Arithmetic

Absolute value is distance. $|a|$ is a with its sign removed: $|3| = 3$ and $|-3| = 3$. So $|a - b|$ is the distance between a and b on the number line, whichever is larger. $|W - \mathbb{E}[W]|$ is the distance between a random variable and its mean.

Squaring removes the sign too. $(-3)^2 = 3^2 = 9$. For any number a , $|a|^2 = a^2$. So $(w - \mu)^2$ is the squared distance between w and μ .

Expanding a square. $(a - b)^2 = a^2 - 2ab + b^2$.

Sums. $\sum_{i=1}^n a_i$ is shorthand for $a_1 + a_2 + \cdots + a_n$. Three rules:

- *Constants factor out.* $\sum_i c a_i = c \sum_i a_i$ when c is the same in every term.
- *A sum of n copies of a constant.* $\sum_{i=1}^n c = nc$.
- *Splitting.* If every term falls into exactly one of two groups, G_1 and G_2 , the sum over all terms equals the sum over the first group plus the sum over the second:

$$\sum_i a_i = \sum_{i \in G_1} a_i + \sum_{i \in G_2} a_i.$$

1.2 Inequalities

1. *Chains.* If $a \geq b$ and $b \geq c$, then $a \geq c$.
2. *Dividing by a positive number keeps the direction.* If $a \geq b$ and $c > 0$, then $a/c \geq b/c$. (Dividing by a negative number reverses it. The proofs below never do that.)
3. *Squaring nonnegative numbers keeps the direction.* If $a > b \geq 0$, then $a^2 > b^2$. For example, $3 > 2$ and $9 > 4$.
4. *Dropping nonnegative terms from a sum cannot increase it.* If every term dropped is zero or positive, what remains is at most the original total.
5. *Shrinking every term shrinks the sum.* If each term of a sum is replaced by a number no larger than it, the new sum is no larger than the old one.

1.3 Limits

“ $a_n \rightarrow 0$ as $n \rightarrow \infty$ ” means a_n gets as close to 0 as you like, and stays that close, once n is large enough. The standard example is $1/n$: it is 0.1 at $n = 10$ and 0.001 at $n = 1000$. The same holds for any fixed number c divided by n : $c/n \rightarrow 0$.

The squeeze rule. If $0 \leq a_n \leq b_n$ for every n , and $b_n \rightarrow 0$, then $a_n \rightarrow 0$. a_n is trapped between 0 and a quantity that shrinks to 0, so a_n shrinks to 0 as well.

1.4 Expectation and variance

W and Y are random variables. a , b and c are constants.

- **Linearity.** $\mathbb{E}[a + bW] = a + b\mathbb{E}[W]$. In particular $\mathbb{E}[c] = c$ and $\mathbb{E}[cW] = c\mathbb{E}[W]$.
- **Sums.** $\mathbb{E}[W + Y] = \mathbb{E}[W] + \mathbb{E}[Y]$, whether or not W and Y are independent. Applied repeatedly, the expectation of a sum is the sum of the expectations: $\mathbb{E}[\sum_i W_i] = \sum_i \mathbb{E}[W_i]$.
- **Scaling.** $\mathbb{V}[cW] = c^2 \mathbb{V}[W]$. Multiplying W by c multiplies every distance from the mean by c . It multiplies every *squared* distance by c^2 .
- **Shifting.** $\mathbb{V}[W + c] = \mathbb{V}[W]$. Adding c moves every value and the mean by the same amount. The distances between them do not change. A constant alone has no variance: $\mathbb{V}[c] = 0$.
- **Variance of a sum.** $\mathbb{V}[W + Y] = \mathbb{V}[W] + \mathbb{V}[Y] + 2\text{Cov}(W, Y)$. For n terms there is one variance per term and one covariance per pair:

$$\mathbb{V}\left[\sum_{i=1}^n W_i\right] = \sum_{i=1}^n \mathbb{V}[W_i] + 2 \sum_i \sum_{j>i} \text{Cov}(W_i, W_j).$$

If the W_i are independent, every covariance is 0.

- **The computational identity.** $\mathbb{V}[W] = \mathbb{E}[W^2] - (\mathbb{E}[W])^2$. Adding $(\mathbb{E}[W])^2$ to both sides gives the form the proofs use:

$$\mathbb{E}[W^2] = \mathbb{V}[W] + (\mathbb{E}[W])^2.$$

1.5 Definitions

The probability of an event is a sum of the PMF. For a discrete random variable W and any set of values A , $\mathbb{P}(W \in A) = \sum_{w \in A} \mathbb{P}(W = w)$. (Distributions handout, section 2.2.)

Finite. A quantity is *finite* when it equals an actual number. A mean or a variance is finite when the average adds up to an actual number instead of growing forever.

Independent and identically distributed. $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} F$ means every X_i has the same distribution F , and the draws are mutually independent. Every X_i then has the same mean μ and the same variance σ^2 .

2. The mean and variance of the sample mean

Throughout this section, $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} F$. Every X_i has the same mean μ and the same variance σ^2 . The sample mean is

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

Each X_i is a random variable, so $\bar{X}$ is a random variable. A new sample gives a new value of $\bar{X}$. Its distribution across repeated samples is its sampling distribution. This section finds the center and the spread of that distribution.

2.1 The expected value

Result. $\mathbb{E}[\bar{X}] = \mu$.

In words: **across repeated samples, the sample means center on the population mean.** $\bar{X}$ is an unbiased estimator of μ (Section 5).

Step 1: the definition.

$$\mathbb{E}[\bar{X}] = \mathbb{E}\left[\frac{1}{n} \sum_i X_i\right].$$

Step 2: move $1/n$ outside. $1/n$ is a constant. By linearity, $\mathbb{E}[cW] = c\mathbb{E}[W]$, with $c = 1/n$ and $W = \sum_i X_i$:

$$\mathbb{E}[\bar{X}] = \frac{1}{n} \mathbb{E}\left[\sum_i X_i\right].$$

Step 3: the expectation of a sum is the sum of the expectations.

$$\mathbb{E}[\bar{X}] = \frac{1}{n} \sum_i \mathbb{E}[X_i].$$

Step 4: use identical distribution. Every X_i has mean μ , so each term of the sum is μ :

$$\mathbb{E}[\bar{X}] = \frac{1}{n} \sum_{i=1}^n \mu.$$

Step 5: add up. A sum of n copies of μ is $n\mu$:

$$\mathbb{E}[\bar{X}] = \frac{1}{n} (n\mu) = \mu. \tag{1}$$

■

The proof used identical distribution in Step 4. It never used independence.

2.2 The variance

Result. $\mathbb{V}[\bar{X}] = \sigma^2/n$.

In words: **the sample means scatter around μ less as the sample grows.** The variance falls in proportion to $1/n$. The standard deviation of $\bar{X}$, $\text{SD}[\bar{X}] = \sigma/\sqrt{n}$, is its standard error. It falls in proportion to $1/\sqrt{n}$. Quadrupling the sample size halves the standard error.

Step 1: the definition.

$$\mathbb{V}[\bar{X}] = \mathbb{V}\left[\frac{1}{n} \sum_i X_i\right].$$

Step 2: move $1/n$ outside. By the scaling rule, $\mathbb{V}[cW] = c^2 \mathbb{V}[W]$. With $c = 1/n$, the constant comes out squared:

$$\mathbb{V}[\bar{X}] = \frac{1}{n^2} \mathbb{V}\left[\sum_i X_i\right].$$

Step 3: use independence. By the variance-of-a-sum rule,

$$\mathbb{V}\left[\sum_i X_i\right] = \sum_i \mathbb{V}[X_i] + 2 \sum_i \sum_{j>i} \text{Cov}(X_i, X_j).$$

The X_i are independent, so every covariance is 0. The second sum drops out:

$$\mathbb{V}[\bar{X}] = \frac{1}{n^2} \sum_i \mathbb{V}[X_i].$$

Step 4: use identical distribution. Every X_i has variance σ^2 :

$$\mathbb{V}[\bar{X}] = \frac{1}{n^2} \sum_{i=1}^n \sigma^2.$$

Step 5: add up and simplify. A sum of n copies of σ^2 is $n\sigma^2$. One factor of n cancels:

$$\mathbb{V}[\bar{X}] = \frac{1}{n^2} (n\sigma^2) = \frac{\sigma^2}{n}. \quad (2)$$

■

The proof used independence in Step 3. If the X_i are correlated, the covariance terms stay. Positive covariances make $\mathbb{V}[\bar{X}]$ larger than σ^2/n . Negative covariances make it smaller.

3. Chebyshev's inequality

Theorem (Chebyshev's inequality). For any random variable W with finite variance and any $\varepsilon > 0$,

$$\mathbb{P}(|W - \mathbb{E}[W]| > \varepsilon) \leq \frac{\mathbb{V}[W]}{\varepsilon^2}.$$

ε (the Greek letter epsilon) is a distance we choose. It can be any positive number. $|W - \mathbb{E}[W]|$ is how far W lands from its mean. The left side is the probability that W lands more than ε away from its mean. The right side is the variance of W divided by ε^2 .

In words: **a random variable with a small variance seldom lands far from its mean.** The inequality puts a number on “seldom.” It uses nothing about the shape of the distribution. The only requirement is a finite variance.

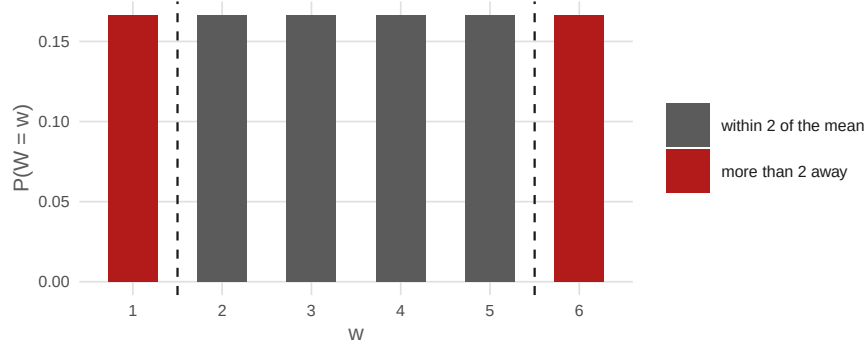
In standard deviations. Let σ be the standard deviation of W , and choose $\varepsilon = k\sigma$ for some $k > 0$. The right side becomes $\sigma^2/(k^2\sigma^2) = 1/k^2$. So for *every* distribution with a finite variance, at most $1/4$ of the probability lies more than 2 standard deviations from the mean. At most $1/9$ lies more than 3 standard deviations away. For a Normal distribution the share beyond 2 standard deviations is about 0.05. Chebyshev's $1/4$ is larger because it must also cover distributions with much heavier tails than the Normal.

3.1 Example: one die roll

Let W be one roll of a fair die. Its mean is 3.5 and its variance is $35/12 \approx 2.92$. Take $\varepsilon = 2$. Only the values 1 and 6 lie more than 2 from 3.5: each is 2.5 away. The value 5, for instance, is 1.5 away. So

$$\mathbb{P}(|W - 3.5| > 2) = \mathbb{P}(W = 1) + \mathbb{P}(W = 6) = \frac{2}{6} \approx 0.33.$$

Chebyshev's bound is $2.92/2^2 = 0.73$. The true probability, 0.33, is below the bound, as the theorem guarantees. The bound is loose: it holds for every distribution with this variance, not only for the die.



Now take $\varepsilon = 1$. The bound is $2.92/1 = 2.92$. Every probability is at most 1, so a bound above 1 carries no information. The bound becomes informative only when ε^2 exceeds the variance, which means ε exceeds the standard deviation.

3.2 Proof

The idea. The variance is a probability-weighted average of squared distances from the mean. Every value more than ε from the mean adds at least ε^2 times its probability to that average. If such values carried a lot of probability, the variance would have to be large. A small variance leaves room for only a little probability far from the mean.

The proof below is for a discrete W . For a continuous W , replace each sum with an integral and each $\mathbb{P}(W = w)$ with $f(w) dw$. Every step then uses the same rule in its integral form.

Write $\mu = \mathbb{E}[W]$ throughout.

Step 1: write the variance as a sum. By definition, $\mathbb{V}[W] = \mathbb{E}[(W - \mu)^2]$. For a discrete random variable, an expectation is a probability-weighted sum over the values, so

$$\mathbb{V}[W] = \sum_w (w - \mu)^2 \mathbb{P}(W = w). \quad (3)$$

Each term is one value's squared distance from the mean, times that value's probability.

Step 2: split the values into two groups. Call a value *far* if $|w - \mu| > \varepsilon$ and *near* if $|w - \mu| \leq \varepsilon$. Every value is in exactly one group. Splitting the sum in (3):

$$\mathbb{V}[W] = \sum_{\text{far } w} (w - \mu)^2 \mathbb{P}(W = w) + \sum_{\text{near } w} (w - \mu)^2 \mathbb{P}(W = w). \quad (4)$$

For the die with $\varepsilon = 2$, the far values are 1 and 6. The near values are 2, 3, 4 and 5.

Step 3: drop the near group. Each term in the near sum is a square, which is at least 0, times a probability, which is also at least 0. So each term is at least 0. By inequality rule 4, dropping the near sum cannot increase the total:

$$\mathbb{V}[W] \geq \sum_{\text{far } w} (w - \mu)^2 \mathbb{P}(W = w). \quad (5)$$

Step 4: replace each squared distance with ε^2 . Every far value has $|w - \mu| > \varepsilon$. Both sides are nonnegative, so by rule 3, squaring keeps the direction: $|w - \mu|^2 > \varepsilon^2$. Since $|w - \mu|^2 = (w - \mu)^2$, every far value has $(w - \mu)^2 > \varepsilon^2$. Multiplying both sides by $\mathbb{P}(W = w) \geq 0$ gives $(w - \mu)^2 \mathbb{P}(W = w) \geq \varepsilon^2 \mathbb{P}(W = w)$.¹ This holds for each term of the far sum, so by rule 5,

$$\sum_{\text{far } w} (w - \mu)^2 \mathbb{P}(W = w) \geq \sum_{\text{far } w} \varepsilon^2 \mathbb{P}(W = w) = \varepsilon^2 \sum_{\text{far } w} \mathbb{P}(W = w). \quad (6)$$

The last equality factors out ε^2 , which is the same in every term.

Step 5: recognize the probability. The far values are exactly the values of W more than ε from μ . The probability of an event is the sum of the PMF over the values in it, so

$$\sum_{\text{far } w} \mathbb{P}(W = w) = \mathbb{P}(|W - \mu| > \varepsilon). \quad (7)$$

Step 6: chain the inequalities. Substitute (7) into (6). Then (5) and (6) link by rule 1:

$$\mathbb{V}[W] \geq \sum_{\text{far } w} (w - \mu)^2 \mathbb{P}(W = w) \geq \varepsilon^2 \mathbb{P}(|W - \mu| > \varepsilon).$$

Dropping the middle term:

$$\mathbb{V}[W] \geq \varepsilon^2 \mathbb{P}(|W - \mu| > \varepsilon). \quad (8)$$

Step 7: divide by ε^2 . $\varepsilon > 0$, so $\varepsilon^2 > 0$. By rule 2, dividing both sides of (8) by ε^2 keeps the direction:

$$\frac{\mathbb{V}[W]}{\varepsilon^2} \geq \mathbb{P}(|W - \mu| > \varepsilon).$$

Written right to left, with $\mu = \mathbb{E}[W]$, this is Chebyshev's inequality. ■

Where the finite variance entered. If the sum in (3) grows forever, $\mathbb{V}[W]$ is infinite. Step 7 then says a probability is at most infinity. Every probability is at most 1, so that statement carries no information.

4. The law of large numbers

4.1 Convergence in probability

Definition (convergence in probability). A sequence of random variables W_n *converges in probability* to a constant c when, for every $\varepsilon > 0$,

$$\mathbb{P}(|W_n - c| > \varepsilon) \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Written $W_n \xrightarrow{p} c$, or $\text{plim } W_n = c$.

¹The strict $>$ becomes $\geq$ here. If $\mathbb{P}(W = w) = 0$, both sides equal 0, and $0 > 0$ is false. $\geq$ covers that case.

$W_1, W_2, W_3, \dots$ is a list of random variables, one for each n . Below, W_n is the mean of a sample of size n . Fix a distance ε . For each n , $\mathbb{P}(|W_n - c| > \varepsilon)$ is one number: the probability that W_n misses c by more than ε . Convergence in probability says this list of numbers goes to zero as n grows.

“For every $\varepsilon > 0$ ” makes the claim hold for every distance, however small. A smaller ε is a stricter standard of “close.” It may take a larger n before a miss becomes unlikely. For every ε , that n exists.

The definition is about probabilities. It does not say W_n equals c for large n . It says a miss by more than ε becomes as unlikely as you like.

4.2 The theorem

Theorem (law of large numbers). If $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} F$ with $\mathbb{E}[X_i] = \mu$ finite, then

$$\bar{X}_n \xrightarrow{p} \mu.$$

The subscript in $\bar{X}_n$ marks the sample size. Each n gives a different random variable, $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Results (1) and (2) from Section 2 hold at every n : $\mathbb{E}[\bar{X}_n] = \mu$ and $\mathbb{V}[\bar{X}_n] = \sigma^2/n$.

The proof below adds an assumption. It requires the variance $\sigma^2 = \mathbb{V}[X_i]$ to be finite. The theorem is true without that assumption. Its proof then needs tools beyond this course. A variable that can only take values in a fixed range, such as a die roll, a turnout indicator or an age, always has a finite variance.

4.3 Proof

The idea. Chebyshev’s inequality bounds the probability of a miss by the variance divided by ε^2 . The variance of the sample mean is σ^2/n , which shrinks as n grows. The bound shrinks to zero. The probability of a miss sits below the bound, so it shrinks to zero too.

Step 1: fix a distance. Let ε be any positive number. Steps 2 to 6 show that $\mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) \rightarrow 0$ for this ε . They use nothing about ε except $\varepsilon > 0$.

Step 2: check that Chebyshev’s inequality applies to $\bar{X}_n$. $\bar{X}_n$ is a random variable. By (2), its variance is σ^2/n . That is finite, because σ^2 is finite and $n \geq 1$. So Chebyshev’s inequality holds with $W = \bar{X}_n$.

Step 3: apply it. With $W = \bar{X}_n$, and $\mathbb{E}[W] = \mu$ from (1), Chebyshev’s inequality reads

$$\mathbb{P}\left(|\bar{X}_n - \mu| > \varepsilon\right) \leq \frac{\mathbb{V}[\bar{X}_n]}{\varepsilon^2}.$$

Step 4: substitute the variance. By (2), $\mathbb{V}[\bar{X}_n] = \sigma^2/n$. Dividing σ^2/n by ε^2 multiplies its denominator by ε^2 :

$$\frac{\mathbb{V}[\bar{X}_n]}{\varepsilon^2} = \frac{\sigma^2/n}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2}.$$

So

$$0 \leq \mathbb{P}\left(|\bar{X}_n - \mu| > \varepsilon\right) \leq \frac{\sigma^2}{n\varepsilon^2}. \quad (9)$$

The left inequality, $0 \leq$, holds because no probability is negative.

Step 5: take the bound to the limit. σ^2 is a fixed number. Step 1 fixed ε . So $c = \sigma^2/\varepsilon^2$ is one fixed number, and the bound is c/n . By the rule for limits, $c/n \rightarrow 0$ as $n \rightarrow \infty$.

Step 6: squeeze. By (9), the probability is at least 0 and at most c/n for every n . By Step 5, $c/n \rightarrow 0$. By the squeeze rule,

$$\mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Step 7: conclude. Step 1 allowed any $\varepsilon > 0$. So the limit in Step 6 holds for every $\varepsilon > 0$. That is the definition of convergence in probability, with $W_n = \bar{X}_n$ and $c = \mu$. Therefore $\bar{X}_n \xrightarrow{p} \mu$. ■

4.4 Example: the mean of n die rolls

For a fair die, $\mu = 3.5$ and $\sigma^2 = 35/12$. Take $\varepsilon = 0.1$. The bound in (9) is

$$\frac{35/12}{n \times 0.1^2} \approx \frac{291.7}{n}.$$

The table compares the bound with the Normal approximation to the actual probability.

n	Chebyshev bound	Normal approximation
100	2.9167	0.5582
1,000	0.2917	0.0641
10,000	0.0292	< 0.0001
100,000	0.0029	< 0.0001

The bound falls in proportion to $1/n$. At $n = 100$ it exceeds 1, so it carries no information. The actual probability is far below the bound at every n . The proof needs only that the bound goes to zero. It does not need the bound to be close.

4.5 Where each assumption was used

Assumption	What it gives	Where it is used
Identically distributed	every X_i has mean μ and variance σ^2	Step 4 of 2.1 and of 2.2
Independent	$\text{Cov}(X_i, X_j) = 0$ for $i \neq j$	Step 3 of 2.2
Finite variance	$\mathbb{V}[\bar{X}_n]$ is an actual number	Step 2 of 4.3

Without independence. Suppose a pollster asks the same person n times, so that every X_i equals X_1 . Then $\bar{X}_n = X_1$ for every n . Its variance is σ^2 at every sample size. The bound in (9) never shrinks. $\bar{X}_n$ does not converge to μ .

Without a finite mean. The Cauchy distribution has no mean. The mean of n independent standard Cauchy draws has the same distribution as a single draw, at every n . Nothing shrinks as n grows.

5. Mean squared error

An **estimand** θ (the Greek letter theta) is the fixed number we want to learn, such as μ . An **estimator** $\hat{\theta}$ is a rule that turns a sample into a guess at θ . Because the sample is random, $\hat{\theta}$ is a random variable.

Definitions. The *bias* of $\hat{\theta}$ is $\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta$. The *mean squared error* of $\hat{\theta}$ is $\text{MSE}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2]$.

$\hat{\theta} - \theta$ is the estimator's error in one sample. The MSE is the average squared error across repeated samples. The bias is the average error, with its sign kept.

Result. $\text{MSE}(\hat{\theta}) = \mathbb{V}[\hat{\theta}] + \text{Bias}(\hat{\theta})^2$.

In words: **an estimator misses in two ways.** Its sampling distribution can be centered in the wrong place, which is bias. It can be spread out, which is variance. The MSE adds the two, with the bias squared so both are in the same units.

5.1 Proof

Step 1: name the error. Let $W = \hat{\theta} - \theta$. W is a random variable because $\hat{\theta}$ is. By definition,

$$\text{MSE}(\hat{\theta}) = \mathbb{E}[W^2].$$

Step 2: use the computational identity. $\mathbb{E}[W^2] = \mathbb{V}[W] + (\mathbb{E}[W])^2$, so

$$\text{MSE}(\hat{\theta}) = \mathbb{V}[\hat{\theta} - \theta] + (\mathbb{E}[\hat{\theta} - \theta])^2.$$

Step 3: simplify the variance. θ is a constant. By the shifting rule, subtracting a constant does not change the variance: $\mathbb{V}[\hat{\theta} - \theta] = \mathbb{V}[\hat{\theta}]$.

Step 4: simplify the expectation. By linearity, with $\mathbb{E}[\theta] = \theta$ because θ is a constant: $\mathbb{E}[\hat{\theta} - \theta] = \mathbb{E}[\hat{\theta}] - \theta$. That is the definition of $\text{Bias}(\hat{\theta})$.

Step 5: substitute Steps 3 and 4 into Step 2.

$$\text{MSE}(\hat{\theta}) = \mathbb{V}[\hat{\theta}] + \text{Bias}(\hat{\theta})^2. \tag{10}$$

■

For an unbiased estimator the bias term is 0, so its MSE equals its variance. By (1) and (2), the sample mean has $\text{MSE}(\bar{X}) = \sigma^2/n$.

5.2 Example: four estimators of a die's mean

Roll a fair die $n = 30$ times. The rolls are iid. Each roll has mean $\mu = 3.5$ and variance $\sigma^2 = 35/12 \approx 2.92$. Suppose we want to estimate μ from the 30 rolls. In this example we know μ , so we can compute each estimator's bias, variance and MSE exactly. Four estimators:

- X_1 , the first roll. $\mathbb{E}[X_1] = \mu$ and $\mathbb{V}[X_1] = \sigma^2$.
- $\frac{1}{2}(X_1 + X_n)$, the average of the first and last rolls. By linearity its mean is $\frac{1}{2}(\mu + \mu) = \mu$. By the scaling rule and independence, its variance is $\frac{1}{4}(\sigma^2 + \sigma^2) = \sigma^2/2$.
- 3, a constant that ignores the rolls. It returns 3 for every sample, so $\mathbb{E}[3] = 3$ and $\mathbb{V}[3] = 0$. Its bias is $3 - 3.5 = -0.5$.
- $\bar{X}$, the sample mean. By (1) and (2), $\mathbb{E}[\bar{X}] = \mu$ and $\mathbb{V}[\bar{X}] = \sigma^2/30$.

By (10), each MSE is the squared bias plus the variance.

Estimator	Bias ²	Variance	MSE
X_1 , the first roll	0.0000	2.9167	2.9167
$\frac{1}{2}(X_1 + X_n)$	0.0000	1.4583	1.4583
3, a constant	0.2500	0.0000	0.2500
$\bar{X}$	0.0000	0.0972	0.0972

The constant 3 is biased, but it has no variance at all. Its MSE, 0.25, is lower than the MSE of the unbiased X_1 . An unbiased estimator is not automatically the more accurate one. $\bar{X}$ has the lowest MSE of the four. Of the three unbiased estimators, $\bar{X}$ has the smallest variance, so it is the most efficient.

6. Why the sample variance divides by $n - 1$

The sample variance from Week 2 divides by $n - 1$, not by n . This section shows that $n - 1$ makes it unbiased for σ^2 . Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} F$ with mean μ and variance σ^2 . Let

$$\text{SSD} = \sum_{i=1}^n (X_i - \bar{X})^2$$

be the sum of squared deviations from the sample mean. Two estimators of σ^2 divide it by different numbers:

$$s^2 = \frac{\text{SSD}}{n - 1}, \quad \hat{\sigma}^2 = \frac{\text{SSD}}{n}.$$

Result. $\mathbb{E}[\text{SSD}] = (n - 1) \sigma^2$. So $\mathbb{E}[s^2] = \sigma^2$, and s^2 is unbiased. $\mathbb{E}[\hat{\sigma}^2] = \sigma^2 - \sigma^2/n$, so $\hat{\sigma}^2$ is biased downward by σ^2/n .

In words: **deviations from $\bar{X}$ run smaller than deviations from μ .** $\bar{X}$ is computed from the same data it is compared against, so it sits closer to the data than the true mean does. Dividing by $n - 1$ instead of n corrects for that shortfall exactly.

6.1 Proof

Step 1: rewrite SSD. Expand each squared deviation with $(a - b)^2 = a^2 - 2ab + b^2$:

$$(X_i - \bar{X})^2 = X_i^2 - 2\bar{X}X_i + \bar{X}^2.$$

Sum over i . Each piece sums separately. $\bar{X}$ is the same in every term, so it factors out of the middle sum. The last piece is n copies of $\bar{X}^2$:

$$\text{SSD} = \sum_i X_i^2 - 2\bar{X} \sum_i X_i + n\bar{X}^2.$$

By the definition of $\bar{X}$, $\sum_i X_i = n\bar{X}$. So the middle term is $2\bar{X} \cdot n\bar{X} = 2n\bar{X}^2$:

$$\text{SSD} = \sum_i X_i^2 - 2n\bar{X}^2 + n\bar{X}^2 = \sum_i X_i^2 - n\bar{X}^2. \quad (11)$$

Step 2: take expectations. Apply the sum rule and linearity to (11). n is a constant:

$$\mathbb{E}[\text{SSD}] = \sum_i \mathbb{E}[X_i^2] - n \mathbb{E}[\bar{X}^2]. \quad (12)$$

Step 3: find $\mathbb{E}[X_i^2]$. By the computational identity, with $W = X_i$, $\mathbb{V}[X_i] = \sigma^2$ and $\mathbb{E}[X_i] = \mu$:

$$\mathbb{E}[X_i^2] = \sigma^2 + \mu^2.$$

Step 4: find $\mathbb{E}[\bar{X}^2]$. By the computational identity, with $W = \bar{X}$, and with (1) and (2):

$$\mathbb{E}[\bar{X}^2] = \mathbb{V}[\bar{X}] + (\mathbb{E}[\bar{X}])^2 = \frac{\sigma^2}{n} + \mu^2.$$

Step 5: substitute Steps 3 and 4 into (12). The first sum is n copies of $\sigma^2 + \mu^2$:

$$\mathbb{E}[\text{SSD}] = n(\sigma^2 + \mu^2) - n \left(\frac{\sigma^2}{n} + \mu^2 \right).$$

Multiply out both brackets:

$$\mathbb{E}[\text{SSD}] = n\sigma^2 + n\mu^2 - \sigma^2 - n\mu^2.$$

The $n\mu^2$ terms cancel:

$$\mathbb{E}[\text{SSD}] = n\sigma^2 - \sigma^2 = (n-1)\sigma^2. \quad (13)$$

Step 6: the two estimators. Divide (13) by each divisor. By linearity, dividing by a constant passes through the expectation:

$$\mathbb{E}[s^2] = \frac{(n-1)\sigma^2}{n-1} = \sigma^2, \quad \mathbb{E}[\hat{\sigma}^2] = \frac{(n-1)\sigma^2}{n} = \sigma^2 - \frac{\sigma^2}{n}.$$

So $\text{Bias}(s^2) = 0$ and $\text{Bias}(\hat{\sigma}^2) = -\sigma^2/n$. ■

Where the missing σ^2 came from. In Step 5, $n\sigma^2$ comes from the X_i and $-\sigma^2$ comes from $n\mathbb{V}[\bar{X}] = n \cdot \sigma^2/n$. The shortfall is the variability of $\bar{X}$ itself around μ . Deviations measured from $\bar{X}$ leave it out. Dividing by $n-1$ puts it back.

As n grows. The bias of $\hat{\sigma}^2$, $-\sigma^2/n$, goes to 0. With a large sample the two divisors give almost the same answer.

Biased but consistent. An estimator $\hat{\theta}_n$ is *consistent* for θ when $\hat{\theta}_n \xrightarrow{P} \theta$ (Section 4.1). $\hat{\sigma}^2$ is biased at every n , but it is consistent: $\hat{\sigma}^2 \xrightarrow{P} \sigma^2$. So is s^2 . The proof applies the law of large numbers to the X_i^2 . It also needs a tool beyond this course, so it is not given here.

7. Functions of the sample mean

We often care about a quantity computed *from* $\bar{X}$ more than about $\bar{X}$ itself. Let g be a rule that turns one number into another, such as squaring it. $\bar{X}$ is a random variable, so $g(\bar{X})$ is a random variable too. A new sample gives a new $\bar{X}$, and with it a new $g(\bar{X})$. This section finds the approximate distribution of $g(\bar{X})$.

The approximation starts from the central limit theorem, as stated in the slides:

Theorem (central limit theorem). If $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} F$ with mean μ and finite variance σ^2 , then as $n \rightarrow \infty$

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1).$$

For large n , $\bar{X}$ is approximately $\mathcal{N}(\mu, \sigma^2/n)$, whatever the shape of F .

$\xrightarrow{d}$ is read “converges in distribution.”

Linear functions are exact. If $g(x) = a + bx$, the rules in Section 1.4 give $\mathbb{E}[g(\bar{X})] = a + b\mu$ and $\mathbb{V}[g(\bar{X})] = b^2\sigma^2/n$. No approximation is needed.

Curved functions are close to a line near μ . Replace g with the straight line that touches it at μ . Call that line’s slope b . Near μ ,

$$g(\bar{X}) \approx g(\mu) + b(\bar{X} - \mu).$$

The right side is a linear function of $\bar{X}$. The rules for linear functions apply to it.

Result (the delta method). For large n ,

$$\mathbb{E}[g(\bar{X})] \approx g(\mu), \quad \mathbb{V}[g(\bar{X})] \approx b^2 \frac{\sigma^2}{n},$$

and $g(\bar{X})$ is approximately $\mathcal{N}(g(\mu), b^2\sigma^2/n)$.

The Normal shape comes from the central limit theorem. A linear function of an approximately Normal variable is approximately Normal.

The line ignores the curvature of g . So for a curved g , these are approximations, not exact results.

Why the approximation improves with n . By the law of large numbers, $\bar{X}$ lands close to μ with high probability. Over that short stretch, the curve and the line nearly coincide.

Sources: Gailmard, *Statistical Modeling and Inference for Social Science*, ch. 7 and section 9.2. Blitzstein and Hwang, *Introduction to Probability*, 2nd ed., sections 10.1 and 10.2. Free at <https://probabilitybook.net/>.